

CHAPTER 12

PERFORMANCE-ENHANCING SUBSTANCES AND METHODS

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After completing this chapter, you will be able to

- provide reliable and up-to-date information to athletes on the risks and benefits of performance-enhancing substances, including anabolic steroids, selective androgen receptor modulators, and peptides;
- understand the efficacy and adverse effects of over-the-counter dietary supplements marketed to athletes for enhancing sport and exercise performance;
- determine which performance-enhancing supplements are beneficial for strength and power performance, aerobic endurance performance, or both; and
- distinguish between those performance-enhancing supplements that mimic the effects of hormones in the body and those that improve performance through some other means.

Athletes who choose to use performance-enhancing substances do so with the hope that they will augment their training adaptations and ultimately improve their sport performance (181). Ideally, performance-enhancing substances also support the health of the athlete and are within the ethical and stated guidelines of the athlete's sport. Because of the ethical considerations relating to unfair advantage during competition and the potential for adverse events, most athletic governing bodies have generated a list of substances that are banned from national and international competition. Athletes caught using such substances can be suspended or forced to forfeit their medals or both. In situations in which the athlete tests positive for a banned substance on repeated occasions, he or she risks a lifetime ban on participation in the given sport. However, numerous nutritional supplements and ergogenic aids are permissible and are frequently used by athletes to maximize performance enhancement. Often the use of these substances is promoted on the basis of unfounded claims (181). Another concern that athletes need to be aware of with the use of dietary supplements is the risk of unintended doping originating from contaminated products. The presence of undeclared compounds in the composition of dietary supplements is a serious concern (131). Thus, it is imperative that the athlete become informed about the legality of these substances, understand the potential risks associated with consumption, and know whether scientific research supports the claims (i.e., the efficacy of the product). The strength and conditioning professional can greatly assist athletes in this regard by giving them relevant information related to these issues, as well as by referring them to nutrition specialists. Although an **ergogenic aid** can be any substance, mechanical aid, or training method that improves sport performance, for the purposes of this chapter the term refers specifically to pharmacologic aids.

Athletes may try to gain a competitive advantage by using supplements that are reputed to be ergogenic but are not banned, or they may knowingly use banned substances in the belief that they

can stay ahead of the drug testers (181). A consequence may be that athletes who would normally refrain from using these substances may feel pressured to use them just to stay abreast of their competitors. However, athletes who are well informed can confidently ignore useless and possibly harmful products despite what their fellow athletes claim. It may also be possible to steer athletes away from the use of banned drugs if they are aware of the risks to their health and safety and are aware that competitors who cheat run a high risk of being detected.

Athletes should focus on using appropriately periodized strength and conditioning methods and sound nutritional practices designed to enhance performance. If these two factors are addressed, at that point the athlete can consider using sport supplements or ergogenic aids. It is important that athletes seek guidance from appropriate professionals to make sure that what they are considering is both legal and efficacious.

An athlete's first priority should be to apply sound principles of training, including adequate nutrition, before using any nutritional supplement or ergogenic aid. Before purchasing or consuming a product, an athlete should seek guidance from qualified professionals to make sure the choice is legal, safe, and effective.

Types of Performance-Enhancing Substances

This chapter discusses two categories of performance-enhancing substances: hormones and the drugs that mimic their effects and dietary supplements. Certain hormones, such as testosterone, play an integral role in the adaptive response to strength and conditioning activities; others, such as epinephrine, are important for energy mobilization during training. These and several other types of hormones are covered more extensively in the next part of this chapter. The distinction between a performance-enhancing drug and a dietary supplement is not intuitively obvious. For example, caffeine, which is found in many beverages such as coffee, is categorized as a drug. The distinction between a drug and a dietary supplement affects whether a product meets United States Food and Drug Administration (FDA) approval for safety and effectiveness. If a product is not classified as a drug or advertised as having therapeutic value, FDA regulations concerning its sale are relatively relaxed. This means that any manufacturer can introduce a new dietary supplement to the market without special approval and that the FDA will not investigate its safety or effectiveness unless a health risk is brought to the agency's attention (76). The FDA's definition of a drug encompasses substances that change the body's structure or function. This includes substances that stimulate hormone secretion. Additionally, if a compound is administered differently from the way in which foods would be consumed, it may be classified as a drug.

The distinction between a drug and a dietary supplement is linked to FDA approval for safety and effectiveness.

Generally, **dietary supplements** are highly refined products that should not be confused with a food. They may not have any positive

nutritional value; hence they are not referred to as *nutritional supplements*. Carbohydrate loading to bolster glycogen stores before an athletic competition is considered sport nutrition, as is a tablet of a single purified amino acid (not promoted for medicinal properties); however, the tablet is considered a dietary supplement.

The FDA regulates both finished dietary supplement products and dietary ingredients. Dietary supplements are regulated under a different set of regulations than those covering conventional foods and drugs (see FDA website at www.fda.gov/Food/Dietarysupplements/default.htm). In 1994, the U.S. Congress passed a seminal piece of legislation known as the Dietary Supplement Health and Education Act (DSHEA). Under DSHEA, manufacturers and distributors of dietary supplements are prohibited from marketing products that are adulterated or misbranded. This means that these firms are responsible for evaluating the safety and labeling of their products before marketing to ensure that they meet all the requirements of DSHEA and FDA regulations. Companies can, however, make claims about effects on the body's structure and function as long as the manufacturers can show that the statements are truthful and not misleading; this is a much less stringent requirement than for effectiveness claims made for drugs.

Definition of Products That Can Be Sold as Dietary Supplements

The following points define which products can be sold as dietary supplements in the United States:

- A product (other than tobacco) intended to supplement the diet that contains one or more of the following dietary ingredients:
 - A vitamin
 - A mineral
 - An herb or other botanical
 - An amino acid
 - A dietary substance for use by humans to supplement the diet by increasing the total dietary intake

- A concentrate, metabolite, constituent, extract, or combination of any ingredient listed above
- The product must also be labeled as a dietary supplement, such that the label must include the term *dietary supplement* or equivalent and therefore cannot also be advertised for use as a conventional food or as the sole item within a meal or diet.

Ergogenic substances are usually banned from athletic competition when a consensus is reached that they may provide an unfair competitive edge or pose a significant health risk. This prohibition does not need to be based on conclusive proof that a substance does anything advantageous; it simply represents an agreement among administrators or clinicians that this may be the case. As mentioned previously, every sport's governing body publishes its own list of banned substances. Probably the most widely recognized international organization regulating doping worldwide is the World Anti-Doping Agency (WADA), which oversees the doping controls and establishes the banned substances list for the International Olympic Committee (61). Each country has an affiliated agency (e.g., United States Anti-Doping Agency [USADA], United Kingdom Anti-Doping [UKAD], and Sport Integrity Australia [SIA]) that oversees anti-doping controls for Olympic sports. Additionally, in some countries these organizations also oversee doping controls for professional sports (i.e., UKAD and SIA). The prohibited substances list is standardized and updated by WADA each year. While the WADA list is the international standard, other organizations, such as collegiate and professional sports in the United States, have different prohibited substances lists as well as penalties for doping. Regardless of which organization regulates doping controls, it is incumbent upon the athlete, sport coach, strength and conditioning professional, and all support staff to ensure that they are compliant with their respective organization's anti-doping policies. [Figure 12.1](#) lists the National Collegiate Athletic Association banned drug classes (updated in 2023). This list is used by many universities in the United States and is subject to change

on a yearly basis. Check with your institution or governing body for the list specific to your situation.

Some of the substances are illegal under governmental statute as well. For example, in the United States anabolic steroids are a Class III substance, which makes their possession, for other than medical use, punishable with a maximum penalty of one year in prison and a minimum US\$1,000 fine if this is the individual's first drug offense. The maximum penalty for trafficking (to deal or trade in something illegal) is no more than a 10-year prison sentence and a fine of no more than US\$500,000 if this is the individual's first felony drug offense. If this is the second felony drug offense, the maximum period of imprisonment and the maximum fine both double. While these penalties are for federal offenses, individual states have also implemented fines and penalties for illegal use of anabolic steroids.

Hormones

A variety of endogenously produced hormones are used to enhance athletic performance. The most commonly used hormone is **testosterone**, along with its synthetic derivatives (115). Testosterone is the primary androgen hormone that interacts with skeletal muscle tissue. In addition to testosterone, a variety of other hormones produced by the body have been employed by athletes as ergogenic aids to stimulate the testes to produce testosterone, or they have anabolic properties in themselves; growth hormone is an example. Erythropoietin, which is secreted by the kidneys, is used to stimulate red cell production for enhancement of aerobic endurance performance, and catecholamines such as adrenaline (or epinephrine), which have metabolic and nervous system effects, are often used to enhance weight loss and to provide greater arousal for performance.

Anabolic Steroids

Anabolic steroids are the synthetic (human-made) derivatives of the male sex hormone testosterone. Physiologically, elevations in testosterone concentrations stimulate protein synthesis, resulting in improvements in muscle size, body mass, and strength (25). Additionally, testosterone and its synthetic derivatives are responsible for the development and maturation of male secondary sex characteristics (i.e., increase in body hair; masculine voice; development of male pattern baldness, libido, sperm production, and aggressiveness). These **androgenic** properties include the full development of the primary sexual characteristics of the male. Thus it is more accurate to refer to synthetic derivatives of testosterone as *anabolic-androgenic steroids*. However, they are also referred to as *androgens*, *androgenic steroids*, or *anabolic steroids*.

Figure 12.1 National Collegiate Athletic

Association banned drug classes.

A. Stimulants

Amphetamine (Adderall), Caffeine (Guarana), Cocaine, Dimethylbutylamine (DMBA; AMP), Dimethylhexylamine (DMHA; Octodrine), Ephedrine, Heptaminol, Hordenine, Lisdexamfetamine (Vyvanse), Methamphetamine, Methylhexanamine (DMAA; Forthane), Methylphenidate (Ritalin), Mephedrone (bath salts), Modafinil, Octopamine, Phenethylamine (PEAs) and its derivatives, Phentermine, Synephrine (bitter orange).

Exceptions: Phenylephrine and Pseudoephedrine are not banned.

B. Anabolic Agents

Androstenedione, Boldenone, Clenbuterol, Clostebol, DHCMT (Oral Turinabol), DHEA, Drostanolone, Epitrenbolone, Etiocholanolone, Methandienone, Methasterone, Nandrolone (19-nortestosterone), Oxandrolone, SARMS [Ligandrol (LGD-4033); Ostarine; RAD140; S-23], Stanozolol, Stenbolone, Testosterone, Trenbolone.

C. Beta-Blockers (Banned for Rifle Only)

Atenolol, Metoprolol, Nadolol, Pindolol, Propranolol, Timolol.

D. Diuretics and Masking Agents

Bumetanide, Canrenone (Spironolactone), Chlorothiazide, Furosemide, Hydrochlorothiazide, Probenecid, Triamterene, Trichlormethiazide.

Exceptions: Finasteride is not banned.

E. Narcotics

Buprenorphine, Dextromoramide, Diamorphine (heroin), Fentanyl and its derivatives, Hydrocodone, Hydromorphone, Meperidine, Methadone, Morphine, Nicomorphine, Oxycodone, Oxymorphone, Pentazocine, Tramadol.

F. Peptide Hormones, Growth Factors, Related Substances, and Mimetics

Growth hormone (hGH), Human Chorionic Gonadotropin (hCG), Erythropoietin (EPO), IGF-1 (colostrum; deer antler velvet), Ibutamoren (MK-677).

Exceptions: Insulin, Synthroid and Forteo are not banned.

G. Hormone and Metabolic Modulators

Anti-Estrogen (Fulvestrant), Aromatase Inhibitors [Anastrozole (Arimidex); ATD (androstatrienedione); Formestane; Letrozole], PPAR-d [GW1516 (Cardarine);

GW0742], SERMS [Clomiphene (Clomid); Raloxifene (Evista); Tamoxifen (Nolvadex)].

H. Beta-2 Agonists

Albuterol, Formoterol, Higenamine, Salbutamol, Salmeterol, Vilanterol.

Note: The banned substance list on [NCAA.org](https://ncaa.org) is updated annually; readers should check the website for a recent update. Also, the list is not exhaustive and only represents some of the banned substances within a particular drug class. Any substance that is chemically/pharmacologically related to one of the above drug classes, even if it is not listed as an example, is also banned. Information about ingredients in medications and nutritional/dietary supplements can be obtained by contacting AXIS at 816-474-7321 or axis.drugfreesport.com.

There is no complete list of banned substances. Do not rely on this list to rule out any label ingredient. Many nutritional/dietary supplements are contaminated with banned substances not listed on the label. It is the student-athlete's responsibility to check with the appropriate athletics staff before using any substance.

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Secretion of testosterone occurs primarily in the interstitial Leydig cells in the testes. Although several other steroid hormones with anabolic-androgenic properties are produced in the testes (e.g., dihydrotestosterone and androstenedione), testosterone is produced in far greater quantities. Testosterone and these other male sex hormones are also secreted in significantly smaller amounts from the adrenal glands (in men and women) and ovaries (in women). Many of the ergogenic aids on the market today are precursors for testosterone (i.e., androstenedione) and are discussed in more detail later in this chapter.

Major Sport Organizations' Lists of Banned Substances

Major League Baseball

<https://img.mlbstatic.com/mlb-images/image/upload/mlb/oqj0ztqxefst9pmkpbcl.pdf>

Major League Soccer

https://democrats-judiciary.house.gov/sites/democrats.judiciary.house.gov/files/mlsconstitution_0.pdf

National Collegiate Athletic Association

https://ncaaorg.s3.amazonaws.com/ssi/substance/2024-25/2024-25NCAA_BannedSubstances.pdf

National Football League

https://nflpaweb.blob.core.windows.net/website/PDFs/List-of-Prohibited-Substances_PES_2022.pdf

National Hockey League

Consistent with World Anti-Doping Agency (WADA) for both in-competition and out-of-competition testing

USA Lacrosse

Consistent with WADA for both in-competition and out-of-competition testing
www.usalacrosse.com/sites/default/files/documents/Safety/2021-World-Lacrosse-Anti-Doping-Code-Final.pdf

USA Powerlifting

Consistent with WADA for both in-competition and out-of-competition testing, with the exception that USA Powerlifting does not enforce
www.usapowerlifting.com/usa-powerlifting-anti-doping-rules/

US Tennis Association

Consistent with WADA for both in-competition and out-of-competition testing and the Tennis Anti-Doping Program
www.usta.com/en/home/about-usta/who-we-are/national/tennis-anti-doping-what-you-need-to-know.html#tab=tennianti-dopingprogramme

World Anti-Doping Agency (WADA)

www.wada-ama.org/en/resources/world-anti-doping-code-and-international-standards/prohibited-list

It was not until the 1930s that testosterone was isolated, synthesized, and subsequently investigated for its effects in humans (24). The physiological changes that testosterone regulates have made it one of the drugs of choice for strength and power athletes or other athletes interested in increasing muscle mass (115). However, testosterone itself is a very poor ergogenic aid. Rapid degradation occurs when testosterone is given either orally or

through injectable administration (24). Thus, chemical modification of testosterone was necessary to slow the degradation process in order to achieve androgenic and anabolic effects at lower concentrations and to provide effective blood concentrations for longer periods of time (24). Many derivatives of testosterone were developed from 1940 to 1960 (186); once these modifications occurred, anabolic steroid use through either oral or injectable administration became possible. In recent years, anabolic steroid administration via creams and gels for topical applications and skin patches has increased in popularity, primarily for medical reasons. However, the most commonly used forms for administration in athletes are oral and injectable forms (115). Examples of oral and injectable anabolic steroids are listed in [table 12.1](#).

Dosing

Athletes typically use anabolic steroids in a **stacking** regimen, in which they administer several different drugs simultaneously (167). The rationale for stacking is to increase the potency of each drug via an additive effect. That is, the potency of one anabolic agent may be enhanced when it is consumed simultaneously with another anabolic agent. Research in this area is limited, and the efficacy of stacking has not been proven one way or the other. Individuals use both oral and injectable compounds. Most users take anabolic steroids in a cyclic pattern, meaning they use the drugs for several weeks or months and alternate these cycles with periods of discontinued use (167). Often users administer the drugs in a pyramid (step-up) pattern in which dosages are steadily increased over several weeks (115). Toward the end of the cycle, the athlete “steps down” to reduce the likelihood of negative side effects. At this point, some athletes discontinue the drug use or perhaps initiate another cycle of different drugs (e.g., drugs that may increase endogenous testosterone production, taken to prevent the undesirable drop in testosterone concentrations that follows the removal of the pharmaceutical agents). One study showed that the typical steroid

regimen involved an average of 3.1 agents, with a typical cycle ranging from 5 to 10 weeks (167). The dose that the athlete administered was reported to vary from between 5 and 29 times greater than physiological replacement doses (167). These higher pharmacologic dosages appear necessary to elicit the gains that the athletes desire. In a classic study on the dose–response curve of anabolic steroids, Forbes (68) demonstrated that the total dose of anabolic steroids has a logarithmic relationship to increases in lean body mass; low doses produce only slight effects, but there is a progressive augmentation of lean body mass with increasingly larger doses. These results reinforce the athlete’s philosophy that if a low dose is effective, more must be better.

TABLE 12.1 Generic and Trade Names of Anabolic Steroids Commonly Used by Athletes

Category and generic name	Example trade names
ORALLY ACTIVE STEROIDS	
Danazol	Danocrine
Ethylestrenol	Maxibolin
Fluoxymesterone	Halotestin Ultandren Android-F
Mesterolone	Proviron Mestranum
Methandrostenolone	Dianabol
Methyltestosterone	Metandren Oreton methyl Android Testered Virolin
Oxandrolone	Anavar

*These are generic categories of substances; many different preparations of each are available.

Category and generic name	Example trade names
	Oxandrin
Oxymetholone	Anadrol-50
Stanozolol	Winstrol

INJECTABLE STEROIDS

Boldenone undecylenate	Equipoise Parenabol
Dromostanolone	Drolban
Methenolone enanthate	Primobolan depot
Nandrolone esters*	Deca-durabolin Durabolin Dynabolan Anabol
Testosterone esters*	Andriol Delatestryl Depo-testosterone Oreton Restandol Sustanon
Trenbolone acetate	Finaject Prabolan

TRANSDERMAL OR TOPICAL STEROIDS

Testosterone	Androderm Androgen Testim Testoderm
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*These are generic categories of substances; many different preparations of each are available.

Athletes typically use higher doses of the substances than are prescribed for men with low testosterone levels. Methandrostenolone (Dianabol), for example, maintains normal secondary sexual characteristics in hypogonadal men at a replacement dose of

approximately 15 milligrams per day; athletes have reported using up to 300 milligrams per day (77). This orally active drug has not been available for medical use in the United States for several decades but is still available through black market sources. Testosterone enanthate is a testosterone ester and is an injectable steroid that is readily available in the United States and is used clinically for some rare diseases and for replacement treatment. A replacement dose is approximately 75 to 100 milligrams per week administered every one to two weeks. Injectable steroids are administered intramuscularly, typically by deep gluteal injections. They are also more potent than oral steroids because of their route of delivery and perhaps also because they do not require additional modification to protect them from immediate metabolism by the liver. The injectable compounds have a wide range of half-lives. Among the testosterone esters, testosterone propionate remains in the circulation for approximately 1.5 days, whereas testosterone buciclate lasts for three months after a single injection (18).

Who Uses Anabolic Steroids?

It is believed that athletes (particularly strength athletes) whose goals are to improve athletic performance are the primary users of anabolic steroids. George J. Mitchell, a former United States senator, in the famous 2007 “Mitchell Report,” stated that use of androgenic-anabolic steroids by Major League Baseball players in the United States was pervasive (41). Before this revelation of anabolic steroid use in athletes, earlier reports date back to the 1952 and 1956 Olympics, in which there were accounts of systematic use of androgens by the Soviet weightlifting team (41). State-sponsored anabolic steroid use in other countries has also been documented. In the former German Democratic Republic after the fall of the communist government in 1990, classified documents revealed a secret state program beginning in 1966 to improve national athletic performance using androgens (41, 75). In the United States, widespread use has also been reported in powerlifters (51), National

Football League players (117), and collegiate athletes (146). Additionally, a 2014 estimate indicated that between 2.9 and 4.0 million Americans have likely used anabolic steroids at some point in their lifetime (173). Evidence also indicates that many steroid users are secretive, with more than 50% of users indicating that they have never disclosed their use of anabolic steroids to their physician (176). Therefore, anabolic steroid use is still prevalent and an important concern among recreational and athletic populations, with some estimates suggesting roughly 100,000 new users per year (24). Although results of several surveys suggest that anabolic steroid use appears to have been declining over the last few decades (213), anabolic steroids are among the top sport issues today because of reported use in many sports over time (73).

Strength athletes are not the only users of anabolic steroids. People outside organized sport use steroids to enhance appearance rather than performance (56). National surveys of male American high school seniors showed that approximately 7% were using, or had used, anabolic steroids (12, 35). One-third of the admitted steroid users were not involved in school-sponsored sports, and more than one-fourth stated that their main reason for using steroids was to improve appearance as opposed to athletic performance. Pope and his colleagues (172, 174) described a subset of bodybuilders with an altered self-image who believed that they looked small and weak even though they were large and muscular. These individuals used ergogenic substances and weight training to increase their body size. Pope calls this condition "reverse anorexia nervosa"; it is also known as **muscle dysmorphia**. These bodybuilders appear to be substantially different from competitive athletes in terms of their objectives, the substantial health risks that some of them are willing to take, and their strategies of using extremely large doses of anabolic steroids. This phenomenon may suggest why the most serious illnesses associated with steroid use have occurred almost exclusively in bodybuilders and not in other steroid-using athletes (77).

Efficacy

The purported ergogenic benefits commonly attributed to anabolic steroid use are increased muscle mass, strength, and athletic performance, particularly in sports requiring maximal strength levels. When anabolic steroids are taken at supraphysiologic doses, these ergogenic benefits are realized (25). The degree and incidence of these changes are variable, depending greatly on, among other factors, the training status of the individual (25).

Muscle Mass and Strength

One of the primary reasons athletes and nonathletes take anabolic steroids is to increase lean muscle mass and maximal strength. If such improvements do occur, this will lead to better performance on the field, with all other variables of athleticism held constant. When anabolic steroids are given in doses similar to those used by recreationally trained and competitive athletes, increases in muscle protein synthesis are seen (98). These increases in protein synthesis are likely responsible for the increases observed in lean body mass in both recreationally trained and competitive athletes taking anabolic steroids (107, 210). Even when anabolic steroids are administered to normal adult men not engaged in intensive resistance training, increases in body mass, including the nonfat component, are observed (25, 69). The extent to which gains in lean body mass and maximal strength occur with anabolic steroid use has been reported in the scientific literature as a case study (3). In this study, an adult male international-level bodybuilder self-injected androgenic hormones (at a dosage of 53 mg/day) for a training period of one year that included only a four-week abstinence from drugs in the middle of the year. During this time, the bodybuilder was able to gain about 7 kilograms (15 lb) of fat-free weight, increase the mean fiber area of the vastus lateralis muscle by approximately 11% in six months' time, and significantly increase maximal strength. Despite these improvements in muscle mass and strength, the individual's health status was negatively affected. Specifically, after drug

withdrawal, the subject experienced atrophic testicles and low luteinizing hormone, follicle-stimulating hormone, and testosterone levels. High-density lipoprotein (HDL) cholesterol was also significantly lowered, which indicates a higher risk for atherogenesis. See more information on the health consequences of anabolic steroid use in the Adverse Effects section.

For some time it was postulated that increases in body mass with androgen use came from an increase in body water (107). An increase in total body water is expected with an increase in muscle mass, since water constitutes a majority of the cellular weight; however, it has been postulated that anabolic steroids may increase water retention as well by increasing interstitial and extracellular volume. Although water retention may explain why not all of the weight gain is sustained after cessation of anabolic steroid use, this issue is still not well understood. In a study of experienced male bodybuilders, an eight-week cycle of nandrolone decanoate (200 mg/week, intramuscularly) resulted in a significant 2.2-kilogram (4.9 lb) increase in body mass, a 2.6-kilogram (5.7 lb) increase in fat-free mass, and a 0.4-kilogram (0.9 lb) decrease in fat mass, with no change in hydration of fat-free mass (210). Additionally, the extracellular and intracellular water ratio was unaltered. Even after six weeks of discontinued androgen use, the body mass of the bodybuilders was still significantly greater than baseline levels (1.6 kg [3.5 lb] greater), yet no hydration changes were seen. The increase in fat-free mass and possible reduction in fat mass may last for several months after cessation of use (69) ([figure 12.2](#)). Thus, athletes may derive a benefit from steroid use even if they stop taking the drugs long enough before competition to obtain a negative drug test. This is why unannounced, year-round drug testing of some elite athletes is important for the prevention of unfair drug use.

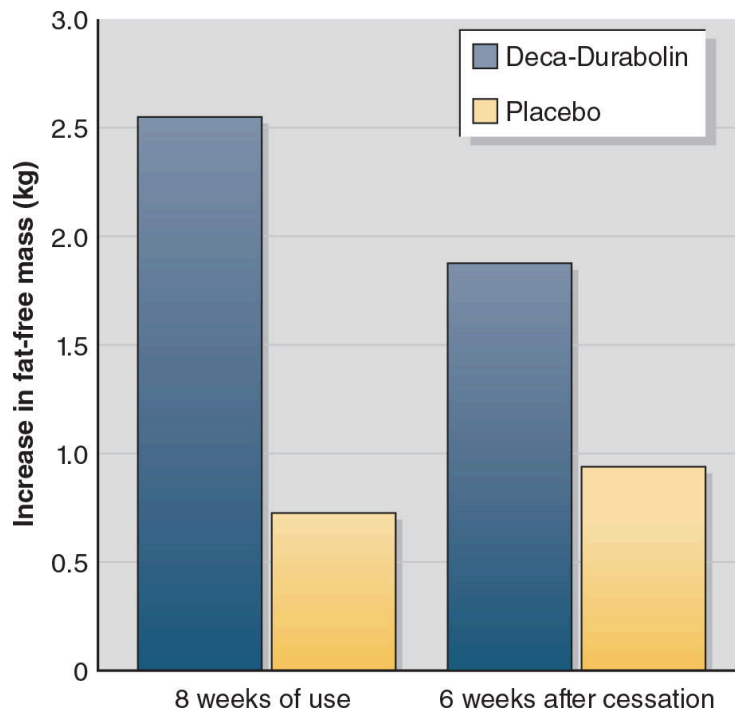


FIGURE 12.2 Changes in lean body mass with anabolic steroid administration and following drug cessation.

Data from Forbes et al. (1992).

Athletic Performance

Initially, researchers examining the ergogenicity of exogenously administered anabolic steroids were unable to see any significant performance effects (74, 201). Consequently, the scientific and medical community suggested that anabolic steroids had little influence on athletic performance. This, however, was contrary to anecdotal reports emanating from gyms and training centers that showed that athletes experienced large improvements in strength in response to using anabolic steroids. On further examination of the initial studies, several methodological flaws became apparent. Several of these studies used physiological doses, in contrast to the suprapharmacologic doses that are typically taken by athletes self-administering androgens. In essence, these subjects were shutting down their own endogenous production and replacing it with an exogenous anabolic steroid. Another flaw in some of these investigations was the method of strength assessment. In several studies, strength performance was evaluated using a mode of

exercise that was different from the training stimulus. This lack of specificity likely masked any possible training effect. Additionally, several studies used subjects who had only minimal resistance training experience (74, 201). When exogenous androgens have been administered to experienced resistance-trained athletes, significant strength gains have been consistently reported (4, 107). Strength gains in experienced strength-trained athletes are generally quite small relative to those seen in novice athletes, but when strength-trained athletes use anabolic steroids, their strength gains may be two to three times higher than those typically seen in similarly trained athletes who are not supplementing (107, 210).

The purported ergogenic benefits commonly attributed to anabolic steroid use are increased muscle mass, strength, and athletic performance, but these changes depend on the training status of the individual.

Psychological Effects

Anabolic steroid use is also associated with changes in aggression, arousal, and irritability (152). The East Germans reportedly used anabolic steroids for this effect, delivering high doses to the central nervous system by taking steroids through the nose. The issue has not been well studied, but anecdotal reports suggest that this practice markedly increased aggressiveness and enhanced performance among athletes (75). Elevations in arousal and self-esteem may be a positive side effect for the anabolic steroid user. Increases in aggressiveness may also be perceived as a benefit, especially for athletes participating in contact sports.

Increased aggressiveness may not be confined to athletic performance, however. Anabolic steroid users experiencing increased aggressiveness may pose a threat both to themselves and to those with whom they come in contact (163, 174). Anabolic steroids are also associated with mood swings and psychotic episodes. Studies have shown that nearly 60% of anabolic steroid users experience

increases in irritability and aggressiveness (174). Pope and colleagues (175) reported significant elevations in aggressiveness and manic scores following 12 weeks of testosterone cypionate injections in a controlled double-blind crossover study. Interestingly, the results of this study were not uniform across the subjects. Most subjects showed little psychological effect, and few developed prominent effects. A cause-and-effect relationship has yet to be identified in anabolic steroid users, but it does appear that individuals who experience psychological or behavioral changes recover when steroid use is discontinued (79).

Adverse Effects

Adverse effects associated with anabolic steroid use are listed in [table 12.2](#). It is important to note that there are differences between the side effects of anabolic steroid use under medical supervision and those associated with abuse (i.e., consumption of many drugs at high doses). Most of the information concerning adverse medical events associated with anabolic steroid use has been acquired from athletes self-administering the drugs. In conjunction, some of the scientific literature has suggested that the medical problems related to anabolic steroids may be somewhat overstated (23, 225), considering that many of the side effects linked to abuse are reversible on cessation. Anecdotally, it appears that a disproportionate magnitude of use and incidence of adverse events are evident in bodybuilders (who are also known for consuming several other drugs—such as diuretics, thyroid hormones, insulin, and anti-estrogens—that relieve some side effects but potentiate other risk factors as well) compared to strength and power athletes.

Testosterone Precursors (Prohormones)

Prohormones are precursors to the synthesis of other hormones and are theorized to increase the body's ability to produce a given specific hormone. The basis for the use of prohormones as an ergogenic aid evolved from a study showing a threefold increase in

testosterone in healthy women who were given 100 milligrams of either androstenedione or dehydroepiandrosterone (143). Athletes who have continued to supplement with these testosterone precursors (androstenedione, androstenediol, and dehydroepiandrosterone [DHEA]) likely do so on the premise that they will increase testosterone concentrations, strength, muscle size, and willingness to train, as well as providing overall athletic performance enhancements similar to those experienced by people taking anabolic steroids. However, these precursors have only relatively weak androgenic properties in themselves; androstenedione and DHEA have only 1/5th and 1/10th the biological activity of testosterone, respectively (153). Nevertheless, testosterone precursors were officially listed as controlled substances in the 2004 Anabolic Steroid Control Act passed by the U.S. Congress, which mandated a physician’s prescription for these substances.

TABLE 12.2 Signs and Symptoms of Ergogenic Aid Abuse

Affected system	Adverse effects
Cardiovascular	Lipid profile changes Elevated blood pressure Decreased myocardial function
Endocrine	Gynecomastia Decreased sperm count Testicular atrophy Impotence and transient infertility
Genitourinary	<i>Males</i> Decreased sperm count Decreased testicular size <i>Females</i> Menstrual cycle irregularities Clitoromegaly Deepening of voice Masculinization

Affected system	Adverse effects
	<i>Males and females</i> Gynecomastia Libido changes
Dermatological	Acne Male pattern baldness
Hepatic	Increased risk of liver tumors and liver damage
Musculoskeletal	Premature epiphyseal plate closure Increased risk of tendon tears Intramuscular abscess
Psychological	Mania Depression Aggression Hostility Mood swings

Designer Anabolic Steroid Control Act of 2014 (DASCA)

In the past few decades, the United States Congress enacted laws regulating anabolic steroids. The first law was the Anabolic Steroid Control Act of 1990. This was updated in 2004. Ten years later, this was updated again as the Designer Anabolic Steroid Control Act of 2014 (DASCA). Noteworthy items about DASCA include the following:

- Congress felt that the laws before DASCA failed to sufficiently curtail the misuse of anabolic steroids.
- Through DASCA, Congress targeted those who would circumvent the Anabolic Steroid Controlled Act of 2004 by manufacturing anabolic steroids with slightly different chemical structures from those expressly controlled but that are intended to cause the same potentially harmful effects.
- DASCA amended the definition of “anabolic steroid” by adding 22 new substances to the prior list.

Studies examining the efficacy of testosterone precursors have produced varying results. No significant differences in strength or body composition were seen in middle-aged men performing a

resistance training program while supplementing with either DHEA, androstenedione (100 mg), or a placebo for three months (219). When DHEA supplementation (150 mg) was examined in a group of younger males (19-29 years) on a two-weeks-on, one-week-off cycle for eight weeks, there were no gains in strength or lean tissue (33). Additionally, the investigators were unable to see any changes in serum testosterone, estrone, estradiol, or lipid concentrations with supplementation. Even in studies using higher dosages (300 mg) of androstenedione for eight weeks, in a similar two-weeks-on, one-week-off protocol, no significant effect was observed in strength, muscle size, or testosterone concentrations (33). However, androstenedione supplementation did cause an increase in serum concentrations of estradiol and estrone and was associated with lower high-density lipoprotein (HDL) levels. These results suggest that although performance changes may not occur in athletes taking this supplement, they may be at higher risk for some of the negative side effects associated with anabolic steroid use. Broeder and colleagues (30), in the conclusion to their Andro Project study in which they investigated the physiological and hormonal influences of androstenedione in conjunction with a high-intensity resistance training program, made the following statement:

Testosterone precursors do not enhance adaptations to resistance training when consumed in dosages recommended by manufacturers. Testosterone precursor supplementation does result in significant increases in estrogen-related compounds, dehydroepiandrosterone sulfate concentrations, down-regulation in testosterone synthesis, and unfavorable alterations in blood lipid and coronary heart disease risk profiles of men aged 35 to 65 years. (p. 3093)

Based on the scientific evidence, it appears as if prohormones fall far short of providing the anabolic effects generally associated with androgens (32). However, scientific research has focused almost exclusively on only a few prohormone supplements—DHEA,

androstenediol, 19-nor-androstenedione, and 19-nor-androstenediol. There are many other prohormone supplements available that have not been clinically tested. Also, use of prohormones has not been studied in highly trained athletes. In addition, oral administration has been the primary method of taking prohormones. The oral use of prohormones may not be as effective as other means of intake (e.g., injection). All of this suggests that continued study of the efficacy of the testosterone precursors for performance enhancement is still needed; specifically warranted is examination of other routes of ingestion in a trained competitive athletic population.

Selective Androgen Receptor Modulators and Peptides

Selective androgen receptor modulators (SARMs) are tissue-selective agonists of androgen receptors in tissues such as muscle and bone. As a result, SARMs are purported to have similar anabolic properties as traditional anabolic steroids but with reduced androgenic effects in other tissues where these effects are undesired (e.g., liver or prostate). Therefore, there is also hope that SARMs may produce more limited side effects when compared to anabolic steroids. The development of novel **peptides**, or short chains of amino acids that regulate cellular functions and biochemical processes, has been an emerging frontier in pharmaceutical development alongside SARMs and, similar to SARMs, has potential for selective agonism and thus limited side effects. Due to their significant clinical potential in diseases or disorders characterized by muscle wasting, SARMs and various peptides have been and continue to be tested in preclinical studies and in clinical trials in various patient populations. However, there are no current FDA-approved SARMs, and all are currently investigational drugs. All SARMs are also currently prohibited both in and outside of competition (i.e., at all times) for all athletes by WADA. Similarly, hormone (e.g., testosterone, growth hormone, corticotrophin)-stimulating peptides are prohibited at all times by WADA. Despite

this, SARMs and peptides are widely marketed to and used recreationally by athletes and have also been found undisclosed in dietary supplements (220). [Table 12.3](#) lists SARMs that are popular among recreational users. Athletes should be aware, however, that products marketed and sold as SARMs or peptides may be just as likely to contain a different drug or no SARMs or peptides at all. Indeed, a chemical analysis of 44 products sold on the Internet as SARMs indicated that only 52% of these actually contained a SARM, 59% contained substance amounts that were different from that listed on the label, and 39% contained a different unapproved drug altogether (212).

TABLE 12.3 Example Selective Androgen Receptor Modulators (SARMs) Popular Among Recreational Users

Research or developmental name	Trade name
MK 2866/GTx-024	Enobosarm Ostarine
GTx-007	Andarine
LGD-4033	Ligandrol
RAD140	Testolone
S-23	—
YK-11	Myanabol
GSK2881078	—

Safety and Efficacy

Similar to anabolic steroids, the purported ergogenic benefits attributed to SARMs include increased muscle mass, strength, and performance. As alluded to previously, SARMs have primarily been tested in the context of clinical trials in patient populations in whom muscle wasting or muscle weakness are a major concern. However, there is some limited available evidence regarding the safety and

efficacy of SARMs in healthy individuals in phase 1 trials. For example, three weeks of LGD-4033 (e.g., ligandrol) use has been shown to promote dose-dependent increases in lean body mass among healthy young and middle-aged men (15). As is often observed with oral androgen administration, these improvements in lean body mass were observed alongside a dose-dependent suppression of total testosterone, sex hormone-binding globulin (SHBG), high-density lipoprotein cholesterol (HDL-C), and triglycerides (15). Similarly, other SARMs such as GSK2881078 and GTx-024 (enobosarm) have been shown to cause dose-dependent increases in lean mass and decreases in HDL-C, SHBG, and total testosterone, suggestive of significant androgen receptor agonism (43, 157), as well as decreased triglycerides, very-low-density lipoprotein, and apolipoprotein A1 (43). Most of these trials also indicate that short-term SARM administration is well tolerated with mild and infrequent adverse events relative to placebo but with consistent decreases in HDL-C and increased alanine transaminase. Thus, concern still exists related to hepatotoxicity with SARM use, and there have been reports of adverse SARM-induced cardiovascular and hepatic outcomes with certain SARMs (101). For example, case reports have associated acute myocarditis and liver injury with RAD140 use by recreational athletes, which apparently resolved with cessation of use (136, 162). Accordingly, the FDA issued a warning to consumers that SARMs have been associated with serious or life-threatening health problems that include increased risk of heart attack or stroke, liver injury, sleep disturbances, sexual dysfunction, and miscarriage.

Several of the peptides that have garnered interest in bodybuilding communities, such as the growth hormone (GH) secretagogue MK-677, showed preliminary efficacy for increasing bone turnover and fat-free mass in older adults (156), but clinical trials were terminated early due to significant safety signals such as increased incidence of congestive heart failure (2). A case report indicated that a five-week cycle of coadministered LGD-4033 and

MK-677 was associated with adverse effects on cholesterol, testosterone, liver function markers, and bone mineral density in a 25-year-old resistance-trained male (37). Somapacitan, an FDA-approved peptide for use in adults with GH deficiency, reduces trunk fat and increases lean body mass in adults with GH deficiency (124). However, it is notable that chronic (~20 weeks) administration of peptides such as hexarelin have no effects on body composition despite causing profound increases in GH release in healthy adults (178). Other peptides, such as the GLP-1 receptor agonists liraglutide and semaglutide, have demonstrated strong efficacy for weight loss in individuals with overweight or obesity (169, 223), and despite a lack of evidence for efficacy among individuals without excess body fat, these weight loss effects may be intriguing to recreational users for cosmetic reasons. [Table 12.4](#) lists some example peptide hormone classes that are explicitly named on the WADA banned list.

Despite the promise of short-term safety and efficacy displayed by several SARMs in initial clinical trials among healthy individuals, substantial work is still necessary to understand the short- and long-term risks associated with SARM and peptide use, understanding that each SARM and peptide is structurally different and is therefore likely to produce different primary and secondary (or side) effects. Moreover, and as noted earlier, in many cases, SARMs available on the market may include other potentially unsafe, nondisclosed substances or may not contain the disclosed SARM at all (212); therefore, risks of SARM or peptide use may be amplified among recreational users. [Figure 12.3](#) lists the short-term changes in lean mass while consuming several different types of SARMs.

TABLE 12.4 Example Peptide Hormone Classes and Specific Peptides Used by Recreational Users Explicitly Named on the WADA Banned List

Testosterone stimulating peptides	Chorionic gonadotropin, luteinizing hormone (LH), gonadotropin releasing hormone (GnRH, gonadorelin) and its agonist analogues (buserelin, deslorelin, goserelin, leuporelin, nafarelin, and triptorelin), kisspeptin and its agonist analogues
Corticotrophin and corticotrophin releasing factors	Coricorelin and tetracosactide
Growth hormone (GH) and its analogues	Lonapegsomatropin, somapacitan, somatogon, AOD-9604, and hGH 176-191
Growth hormone secretagogues/releasing peptides	GHRH, sermorelin, CJC-1293/1295, tesamorelin, MK-677, lenomorelin (ghrelin), anamorelin, macimorelin, tabimorelin, capromorelin, alexamorelin, examorelin (hexarelin), GHRP-1–6
Growth factors and growth factor modulators	IGF-1, mecasermin, FGFs, HGF, MGFs, PDGF, TB-500, VEG-Fs, and any other growth factors or growth factor modulators
Erythropoietins (EPO) and agents affecting erythropoiesis	Erythropoietin receptor agonists (dEPO), EPO, EPO-based constructs (EPO-Fc, CERA), EPO-mimetic agents and their constructs (CNTO-530, peginesatide); hypoxia inducible factor (HIF) activating agents (cobalt, daprodustat, IOX2, molidustat, roxadustate, vadadustat, and xenon); GATA inhibitors; transforming growth factor beta (TGF- β) signaling inhibitors (luspatercept, sotatercept); innate repair receptor agonists (asialo EPO, CEPO)

Data from World Anti-Doping Agency, "World Anti-Doping Code International Standard Prohibited List 2025," last modified September 12, 2024, https://www.wada-ama.org/sites/default/files/2024-09/2025list_en_final_clean_12_september_2024.pdf.

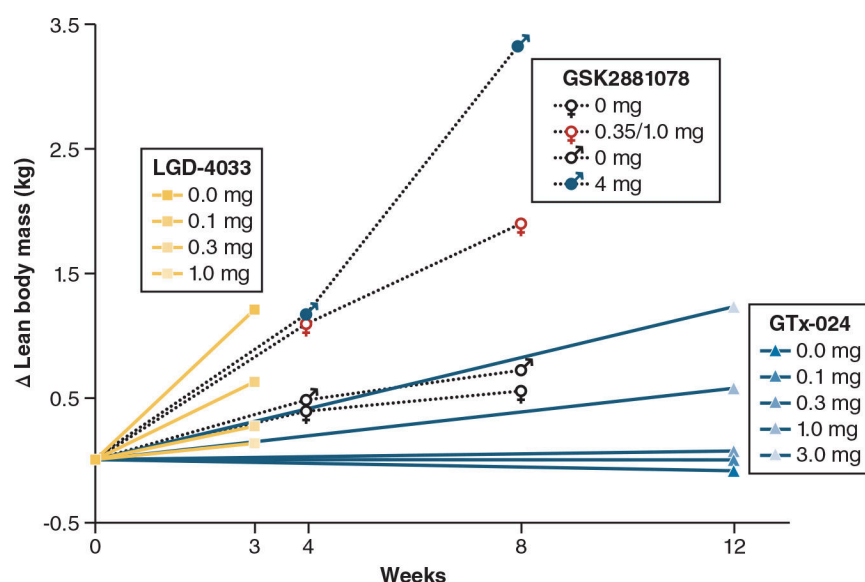


FIGURE 12.3 Changes in total body lean mass in response to short-term LGD-4033 administration in healthy young men (squares), GSK2881078 administration in older men and postmenopausal women (gender symbols), and GTx-024 in older men and women (triangles). Changes in lean body mass generally demonstrated dose dependency, with greater changes observed in response to higher doses.

Data from Basaria et al. (2013), Neil et al. (2018); Dalton et al. (2011).

Adverse Effects

No known reports or scientific studies have investigated the adverse effects of selective androgen receptor modulators and peptides.

HCG (Human Chorionic Gonadotropin)

Human chorionic gonadotropin (HCG) is a hormone obtained from the placenta of pregnant women and is very closely related in structure and function to luteinizing hormone. In fact, it is the pregnancy indicator used by over-the-counter pregnancy test kits, since it is generally not found in the body at any other time. In the general population, HCG is sometimes injected into overweight females under medical supervision in conjunction with low-calorie diets for the purposes of weight loss (87). Research suggests that when used for this purpose, HCG is not effective for inducing weight loss. The caloric restriction that accompanies the HCG injections appears to be the main contributing factor to associated weight loss (87).

Efficacy

While HCG offers females no performance-enhancing ability, it is cited anecdotally as useful for males who take anabolic steroids. When injected into men, HCG can increase testicular testosterone production; testosterone levels can nearly double within four days after a large intramuscular injection (48). The activity of HCG in the male body is due to its ability to mimic luteinizing hormone, a pituitary hormone that stimulates the Leydig cells in the testicles to manufacture testosterone. The reason males would want to be injected with HCG to increase endogenous testosterone levels is that endogenous testosterone production is suppressed at the end of a steroid cycle (151, 154). For that reason, if HCG is used by athletes, it is likely used by those who are finishing a cycle of anabolic steroids and are looking to activate their own endogenous testosterone production.

Adverse Effects

Human chorionic gonadotropin is given as an injection under the skin or into a muscle; the side effects that are common to such administration are pain, swelling, and tenderness around the injection site. There is very little research on the side effects of HCG injections. In one investigation in which HCG was injected into obese women, no adverse effects were reported in relation to blood pressure or routine blood work (197).

Insulin

Insulin is a potent anabolic hormone. It is secreted by the pancreas in response to elevations in blood glucose or specific amino acid (e.g., leucine) concentrations. Its role is to facilitate the uptake of glucose and amino acids into the cell. Since insulin increases protein synthesis, it is considered an anabolic hormone.

Efficacy

When insulin concentrations are elevated naturally (i.e., endogenously from the pancreas following carbohydrate ingestion), there are no safety concerns in otherwise healthy individuals. Additionally, it has been reported that postworkout carbohydrate ingestion suppresses muscle protein breakdown via the anticatabolic effects of insulin (29). Theoretically, if protein breakdown is suppressed over several weeks to months, gains in lean muscle mass could be realized.

Adverse Effects

Because of insulin's anabolic properties and its rumored ability to potentiate the effects of growth hormone and insulin-like growth factors, some individuals (primarily bodybuilders) take insulin by injection. The use of insulin in this manner comes with serious consequences. Possible outcomes in a previously healthy athlete include immediate death or coma due to severe hypoglycemia (106).

Human Growth Hormone

Human growth hormone (HGH), a protein secreted from the anterior pituitary gland, has several important physiological functions that enhance its ergogenic effect. It is anabolic due to its stimulation of bone and skeletal muscle growth, but it also has important metabolic functions such as maintaining blood glucose levels, increasing the uptake of glucose and amino acids into muscle cells (67), and stimulating release of fatty acids from the fat cells.

The primary source of pharmacologic growth hormone is a relatively complicated molecule from which it is synthetically derived using recombinant DNA technology. However, until 1986, the only source of the hormone was the pituitaries of human cadavers. Because HGH receptors are unable to cross-react with growth hormone from animal sources, the financial cost of HGH was very high before 1986. Although the use of cadaver growth hormone came with significant health risks, these consequences did not

prevent athletes from taking the supplement but instead only substantially restricted use (115). The development of recombinant HGH provided clinicians with a relatively low-risk drug at a lower cost (as compared to cadaver growth hormone) with greater availability. Clinicians can now prescribe recombinant HGH to increase stature in short children with growth hormone deficiency and alter body composition in adults. The use of HGH as a performance-enhancing substance is believed to be widespread among professional athletes; it is either taken alone or stacked with anabolic steroids (16). Even though recombinant technology has increased the availability of HGH, its cost, especially on the black market, is extremely high—between several hundred and a few thousand dollars per month. Humatrope, Nutropin, Norditropin, Genotropin, Omnitrope, Serostim, Saizen, and Protropin are brand names of HGH commonly used in the United States.

Efficacy

There appear to be no studies on the efficacy of HGH in professional athletes. Most investigations of HGH have focused on HGH as replacement therapy in growth hormone-deficient adults and children or in healthy older individuals. These studies have consistently shown positive alterations in body composition (increases in lean body tissue with decreases in body fat) (66, 204). In men with established growth hormone deficiencies, nightly injections of recombinant HGH for six months resulted in an average increase of 12 pounds (5.4 kg) in lean body mass and a similar amount of fat loss (184). Most studies have not addressed the effect of HGH therapy on muscle strength and performance. One study showed no changes in isokinetic strength following 12 months of therapy (112). However, the subjects in that study did not perform any resistance training during the course of treatment. A study involving trained adults given growth hormone (three days per week for six weeks) showed modest changes in body composition, but no strength assessment was performed (50). Although the scientific

literature does not provide support for the efficacy of HGH use in athletic populations, it is likely that the inability to perform such studies (due to ethical constraints) will limit much of our understanding of HGH and human performance. In years past, the use of HGH by athletes may have been prevalent in some professional sports because of its perceived efficacy and the fact that it could not be detected in **random drug tests**. Currently, HGH cannot be detected in the urine via a drug test, and it may be postulated that athletes opt to use it for this reason. However, a blood test for HGH was first introduced at the 2004 Summer Olympic Games in Athens, Greece.

Growth hormone is a protein molecule. Injection is necessary in order to avoid its complete metabolism and maintain its effectiveness. Oral ingestion does not result in any benefit. Many of the actions of growth hormone are mediated through insulin-like growth factor-1 (IGF-1), another peptide hormone, which is produced and secreted from the liver in response to growth hormone stimulation. IGF-1 is now being synthesized using recombinant DNA technology and will likely produce the same effects as HGH.

Adverse Effects

The use of HGH does, however, present some potential significant health risks. Excessive secretion of growth hormone during childhood causes gigantism, a condition in which a person becomes abnormally tall. After puberty, once linear growth has stopped, excess secretion of growth hormone causes **acromegaly**, a disfiguring disease characterized by a widening of the bones, arthritis, organ enlargement, and metabolic abnormalities. This is a potential risk for athletes who use HGH as an ergogenic aid. Additionally, these side effects may provide some indication that the athlete could be using this drug. In clinical studies in growth hormone-deficient adults, side effects appear to be minimal for even up to two years of replacement therapy (1, 209). However, athletes who supplement with HGH generally use dosages that far exceed the

doses commonly administered in replacement treatment (16). Thus, one should not assume that HGH use is benign with regard to adverse medical events in doses commonly used by an athletic population. Adverse side effects for HGH abuse are diabetes in prone individuals; cardiovascular dysfunction; muscle, joint, and bone pain; hypertension; abnormal growth of organs; and accelerated osteoarthritis (16).

Although growth hormone used as replacement therapy for people with growth hormone or IGF-1 deficiency can be effective and can have minimal adverse consequences, the dosages that are likely used by athletes may pose a significant risk for acromegaly.

Erythropoietin

One of the limiting factors for endurance performance is the athlete's ability to deliver oxygen to the contracting skeletal muscle. Over the years, several methods have been developed to increase the body's ability to deliver oxygen. Some methods include novel training programs that may naturally elevate red blood cell and hemoglobin levels (i.e., living and training at altitude), while other methods involve synthetic means to elevate the oxygen-carrying capacity of the blood (i.e., blood doping). Blood doping increases one's red blood cell mass and is typically conducted using blood transfusions or taking a hormone known as *erythropoietin*. There are two forms of blood doping via blood transfusions: autologous and homologous. Autologous blood doping is the transfusion of one's own blood, which has been stored (refrigerated or frozen) until needed. Homologous blood doping is the transfusion of blood that has been taken from another person with the same blood type. Since the late 1980s, blood doping via autologous blood transfusion has taken a back seat to the more popular method, administration of recombinant human erythropoietin (39).

Erythropoietin (EPO) is produced in the kidneys and stimulates the production of new red blood cells. It is also a protein hormone that can be produced by recombinant DNA techniques and is reportedly being abused by athletes (64). Erythropoietin use was one of the methods of doping that Lance Armstrong admitted to over the course of his cycling career. The EPO level in the blood increases in response to chronic aerobic endurance exercise. In certain types of anemia, especially in kidney patients with inadequate EPO production, recombinant human EPO can improve the quality of life of the individual.

Efficacy

Injections of EPO are generally associated with elevations in both hematocrit and hemoglobin. A 2013 systematic review and meta-analysis indicated that EPO administration promotes increases in hemoglobin and hematocrit of 5% to 17% and 8% to 19%, respectively (108). These increases are accompanied by increases in aerobic capacity and maximal power output of 7% to 10%, which are in turn linked with improvements in time to exhaustion in open endpoint exercise for both trained (9%-17%) and untrained (22%-50%) individuals (108). Therefore, the ability of EPO to enhance oxygen-carrying capacity of the blood suggests it is likely to be an effective ergogenic aid for the aerobic endurance athlete. Initial evidence also indicates that 16 days of EPO administration can elicit changes in the skeletal muscle proteome in a manner that supports aerobic exercise performance (e.g., increased oxidative enzyme activity) (42). However, it is also important to note that the effect of EPO administration on time trial performance or other parameters critical toward aerobic exercise performance such as the ventilatory threshold, critical velocity, or critical power is not well understood. While much of the research conducted with EPO is focused on endurance performance, there is theoretical rationale for EPO use by anaerobic athletes to improve recovery from high-intensity efforts.

Adverse Effects

While medically supervised use of EPO provides therapeutic benefit in the treatment of anemia related to kidney disease, its misuse can lead to serious health risks for athletes who use this substance simply to gain a competitive edge. The increase in hematocrit resulting from EPO injection presents a significant health risk. Increases in red blood cell number increase blood viscosity (the thickening of the blood). This poses several problems that include increased risk of blood clotting, elevations in systolic blood pressure and accentuation of the exercise pressor response stroke, and cerebral or pulmonary embolism (22, 42, 82). During aerobic endurance events, the additional problem of dehydration could compound cardiovascular risks by eliminating any safety margin in the balance between performance advantages from artificially increased hematocrit and decrements from increased blood viscosity. Several studies have noted marked accentuation of the blood pressure response during exercise following EPO administration. The deaths of a number of competitive cyclists have been related to EPO administration (82). The primary risk associated with EPO is its lack of predictability compared to red blood cell infusion. Once EPO is injected into the body, the stimulus for producing red blood cells is no longer under control. Consequently, aerobic endurance athletes should avoid this drug because of the significant cardiovascular risk, leading to possible death, associated with its administration.

β -Adrenergic Agonists

Synthetic β -adrenergic agonists, or **β -agonists**, are substances chemically related to epinephrine, a hormone produced in the adrenal medulla that regulates physiological effects, such as **lipolysis** (the breakdown of fat) and **thermogenesis** (increased energy expenditure resulting in the production of heat). β -Agonists were originally developed for the treatment of asthma and other life-threatening medical conditions. Some of these compounds have been found to have specific effects on body composition, such as

increases in lean mass and decreases in stored fat (177); because of this, these drugs are sometimes referred to as *partitioning agents*. One of the more popular β -agonists used by athletes is the drug clenbuterol (177).

Efficacy

Clenbuterol is a β_2 -agonist and is a widely used bronchodilator (used to reverse bronchial restriction) in many parts of the world. Athletes use clenbuterol as an ergogenic aid to increase lean muscle tissue and reduce subcutaneous fat (177). Data on the effectiveness of clenbuterol have typically been based not on healthy athletes, but rather subjects with heart failure (125) and patients experiencing muscle-wasting conditions (144). Though studies in humans are limited, several findings have indicated an ergogenic potential of β_2 -agonists for strength improvements (145). Athletes generally use clenbuterol in doses twice the recommended amounts administered for clinical purposes, in a cyclic fashion (three weeks on alternated with three weeks off, with a two-days-on, two-days-off cycle during the “on” week) (177).

Adverse Effects

Athletes consume clenbuterol in a capsule form, in contrast to the inhalation route that is often used for relieving bronchial constriction. Although a number of potential side effects have been suggested (i.e., transient tachycardia, hyperthermia, tremors, dizziness, palpitations, and insomnia), actual documented events are quite limited (196). In addition, the scarcity of data on the ergogenic potential of clenbuterol in humans makes determining its efficacy difficult.

β -Blockers

β -Blockers are a class of drugs that block the β -adrenergic receptors, preventing the catecholamines (i.e., norepinephrine and epinephrine) from binding. β -Blockers are generally prescribed by

cardiologists for the treatment of a wide variety of cardiovascular diseases, including hypertension. The ergogenic benefit of these drugs may reside in their ability to reduce anxiety and tremors during performance (138). Thus, athletes who rely on steady, controlled movements during performance (i.e., archers or marksmen) would appear to benefit from these drugs. Also, β -blockers may improve physiological adaptations from aerobic endurance training by causing an upregulation of β -receptors. If this is true, it would result in an exaggerated response to sympathetic discharge during intense exercise on cessation of supplementation.

Efficacy

Several studies have shown that β -blockers can improve both slow and fast shooting accuracy (7, 135). In addition, the dose taken appears to have significant effects on the magnitude of improvement. In shooters who administered β -blockers in two different doses (80 mg vs. 40 mg of oxprenolol), the group taking the higher dose shot with greater accuracy (7). In certain sports, however, some degree of anxiety may be important. Tesch (205) reported that bowlers whose performance was improved during blockade with oxprenolol had significantly greater heart rates before, during, and after competition than subjects whose performance did not improve while on β -blockers.

Impact of Alcohol on Performance

Both acute and chronic alcohol consumption can impede exercise performance by adversely affecting several physiological processes critical toward optimizing performance and recovery. Perhaps most obviously, acute alcohol consumption can impair performance in skilled tasks by effects on fine motor control, reaction time, judgment, and arousal (191). However, acute alcohol consumption has also been shown to impede skeletal muscle metabolism, recovery, and adaptation. For example, isometric and dynamic strength loss induced by strenuous eccentric exercise is exacerbated by high-dose postexercise alcohol consumption (~6.3 standard drinks over 1.5 hours, where a standard drink contains 14 g of pure alcohol) in young men (14), although strength trained women may be protected from this effect (137). Similarly, mechanistic studies suggest that acute high-dose